

NIGUSS GEBRU

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EDUCATION

Georgia State University Perimeter College

Associate of Science in Computer Science

Clarkston, GA

Expected May 2027

Honors College | GPA: 4.0/4.0

President's List (two consecutive semesters), Phi Theta Kappa (PTK) Honor Society, MESA Scholar (2026), CLACEC Annual Coding Competition Winner (2026), ColorStack Member, Research Scholar

Relevant Coursework: Programming Principles, Computer Science I & II (Honors), Data Structures & Algorithms, Web Development, Database Management Systems, Principles of Physics I, Calculus I & II

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C++, C#, PHP, Go, Dart, Kotlin, R, SQL, MATLAB, HTML, CSS

Frameworks/Libraries: React.js, Next.js, Node.js, Express.js, Django, FastAPI, Flask, React Native, Bootstrap, Pandas, scikit-learn, Matplotlib, OpenCV, Google ADK, JUnit

Tools/Databases: Git/GitHub, Docker, Kubernetes, PostgreSQL, MySQL, Firebase, Supabase, Google Cloud, SQLite, Sequelize, Drizzle, Swagger/OpenAPI, REST API, Google App Engine, Cloudflare, Figma, Jupyter, VS Code, Unity, Qiskit, Linux

PROJECTS

Fresh Land (Immigrant Onboarding Platform) – Founder

Dec 2025 – Present

- Architected and deployed a production web app (freshland.cc) using Next.js and TypeScript, featuring 762+ verified Georgia resource directory, a Reddit-style community forum, and a structured newcomer pathway with 10+ stateful steps and per-user completion tracking, currently serving 150+ daily active users.
- Engineered and optimized PostgreSQL backend with 16+ relational tables, 14+ versioned schema migrations and RLS policies enforcing per-user authorization at the database layer; built a multilingual AI assistant by integrating Anthropic and Gemini APIs with custom server-side inference pipelines and real-time token streaming, reducing user response time by 60%.

IoT Attendance System – New English Private School (NEPS)

Oct 2023 – Feb 2024

- Co-engineered and deployed a full-stack IoT attendance system using C++, replacing paper-based records with automated RFID card scanning and Arduino powered biometric verification, achieving a 91% reduction in manual attendance logging time and reducing transcription errors by 35% through real-time, data-driven syncing to a MySQL and Firebase backend.
- Designed the system to support concurrent multi-reader deployments across campus with low-latency C++ data capture, enabling admins to instantly query and view attendance records across all departments from a centralized dashboard.

Front-End Developer – New English Private School (NEPS) Website

April 2023 – May 2024

- Engineered the full frontend of the school's official website within a 4-person team using HTML5, CSS3, and vanilla JavaScript, implementing interactive menus, sliders, animations, and dynamic navigation features that modernized the school's digital presence.
- Integrated jQuery-powered UI components across multiple pages ensuring seamless frontend and backend connectivity with Node.js and Firebase APIs, scaling the site to 75,000+ monthly visitors school-wide.

Robotic Assistive Arm – New English Private School (NEPS)

Jan 2023 – May 2023

- Led a 3-member team to design and build a 6-DOF robotic assistive arm using a Raspberry Pi for high-level control and an Arduino ESP32 programmed in C++ for real-time coordination of six servo motors, motion sequencing, and safety I/O.
- Integrated a robotic gripper, joint mechanisms, and embedded control systems into a functional feeding assistant, improving mealtime independence for individuals with limited mobility and earning 1st Place at the Addis Ababa district-level competition.

PROFESSIONAL EXPERIENCE

Undergraduate Researcher – Center for Outreach, Research and Engagement (C.O.R.E.)

Georgia State University Perimeter College (Under the Supervision of Dr. Andrew Kim)

May 2026 – Present

Remote/Dunwoody, GA

- Developed a full stack web-based Computerized Adaptive Testing (CAT) platform featuring a novel three-question block algorithm that dynamically adjusts question difficulty based on student performance, designed with psychological comfort principles to minimize test anxiety during assessment.

- Administered a compensated multi-institutional survey to 135+ students across diverse academic backgrounds to empirically validate a five-tier difficulty classification system using PHP and MySQL, applying a data-driven Item Response Theory (Rasch Model) approach to calibrate question difficulty parameters and improve classification accuracy for personalized adaptive assessment.

Supplemental Instructor

Georgia State University Perimeter College

May 2026 – Aug 2026

Dunwoody, GA

- Collaborated with two instructors to improve student success by designing and executing weekly academic support plans for Precalculus (MATH 1113), boosting top-grade attainment by 73%, a marked improvement over the two preceding semesters.
- Delivered 160+ hours of targeted instruction through office hours, assignment support, and exam review sessions tailored to diverse learning needs.

Student Assistant Lead – Student Life Department

Student Engagement Center, Georgia State University Perimeter College

July 2026 – Present

Clarkston, GA

- Spearheaded recruitment and onboarding for Student Life, designing comprehensive training programs for high-performing volunteer teams.
- Architected campus-wide events from conceptualization to execution, leveraging creative design to maximize student participation and build an inclusive campus community.

Private Tutor

Superprof Platform (Online and In-Person)

Dec 2024 – Present

Snellville, GA

- Delivered personalized instruction in AP Algebra, AP Pre-Calculus, AP Calculus, AP Statistics, and AP Physics to high school students through one-on-one and small-group sessions.
- Organized large-scale midterm and final exam reviews with 25+ attendees; regular participants consistently ranked in the top tier of their class.

LEADERSHIP & EXTRACURRICULARS

Speaker of the Senate – Student Government Association (SGA)

Georgia State University Perimeter College

July 2026 – Present

Clarkston, GA

- Serving as the highest-ranking legislative officer of the Campus Senate, presiding over all senate proceedings, presiding over all Campus Senate proceedings and legislation while championing student representation and driving meaningful policy changes that directly impact the Perimeter College.

Event Organizer – Clarkston Computing and Engineering Club

Georgia State University Perimeter College

Aug 2026 – Present

Clarkston, GA

- Planned and managed events including the Annual Boat Regatta, robotics workshops, bridge-building contests, and coding challenges, averaging 25 participants per event.
- Managed the club's event budget and logistics, contributing to the organization's strong community presence on campus.

STEM Program Coordinator – Technology Student Association (TSA)

Brookwood High School

Aug 2024 – May 2025

Snellville, GA

- Facilitated weekly STEM workshops for 25+ teams, promoting innovation and collaborative project design.
- Led a cross-functional team at GSMST for Gwinnett Hackathon IV to build a Unity-based educational game, overseeing C# scripting, asset creation, and version control.

ICT Instructor Assistant

New English Private School

Dec 2023 – May 2024

Addis Ababa, Ethiopia

- Directed ICT classes for 40+ students, improving programming literacy and engagement by 30% through Python and web development curricula.
- Led project-based learning sessions emphasizing teamwork and real-world problem-solving in technical environments.

VOLUNTEER EXPERIENCE

President, Charity Club – SOS Children's Villages

New English Private School

Mar 2023 – Feb 2024

Addis Ababa, Ethiopia

- Led a student-run charity group supporting SOS Children's Villages, raising \$1,250+ in donations to provide food, groceries, and clothing for children in need.
- Organized donation drives and community outreach efforts, coordinating the full group to ensure sustainable support for 12+ months.